4 **(a)** The electric potential at a point in an electric field is defined as the work done per unit positive charge by an external force
in bringing a small test charge from infinity to that point.

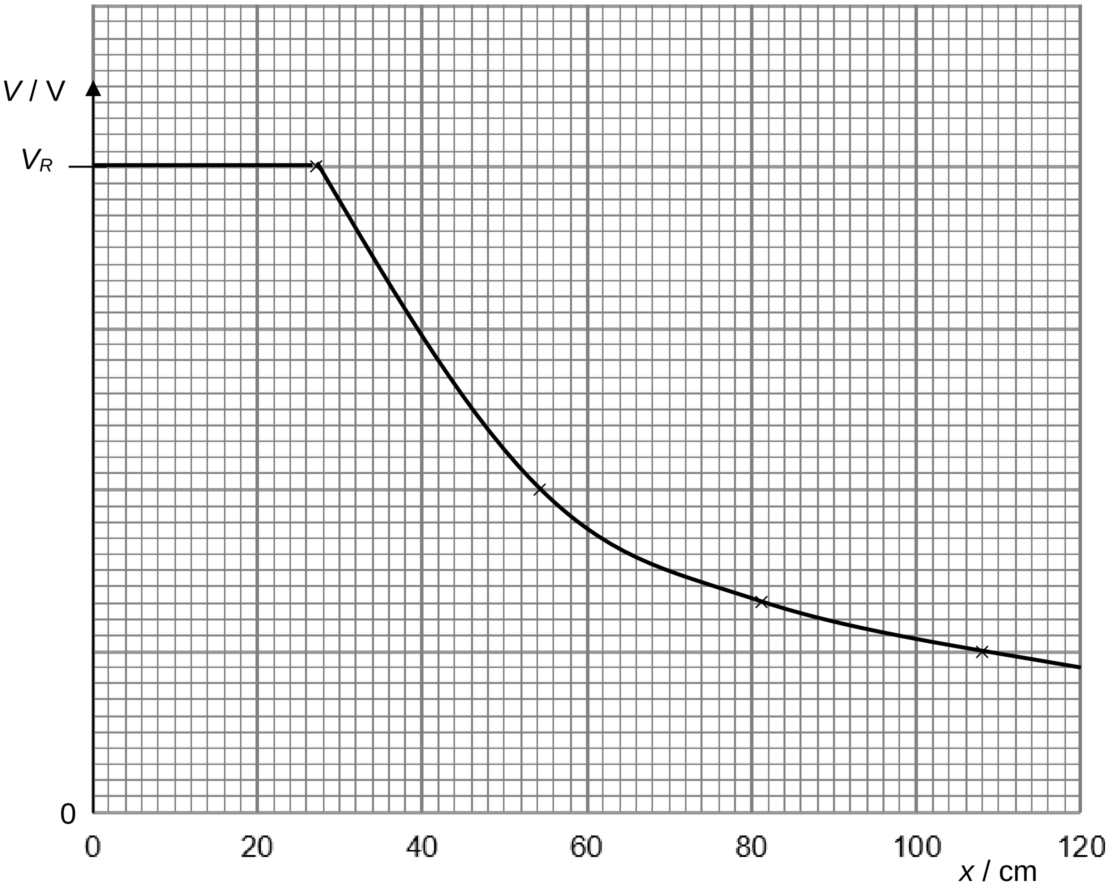
(b) **(i)**

$$V_R = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{0.060\times10^{-6}}{0.27}$$

$$= 2.00\times10^{3}\ \text{V}$$

(ii)



At x = 27 cm, V is 40 small squares. Using $V \propto \frac{1}{r}$,

When x is 2 × 27 = 54 cm, V is 20 small squares.
When x is 3 × 27 = 81 cm, V is 13.3 small squares.
When x is 4 × 27 = 108 cm, V is 10 small squares.

(iii) The directions of E_A and E_B are both towards the right, in the same direction.

$$E_A = \frac{1}{4\pi\varepsilon_0}\frac{Q_A}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{0.060\times10^{-6}}{0.70^2} = 1.101\times10^{3}\ \text{N C}^{-1}$$

$$E_B = \frac{1}{4\pi\varepsilon_0}\frac{Q_B}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{0.035\times10^{-6}}{0.70^2} = 6.423\times10^{2}\ \text{N C}^{-1}$$

$E = E_A + E_B = 1.101 \times 10^3 + 6.423 \times 10^2$
$= 1.74 \times 10^3$ N C^{-1}

(iv) The charges on sphere A will redistribute such that there are more positive charges on the right side of sphere A.
Hence, the electric field strength due to sphere A will be greater in magnitude.

5 **(a)** $Q = It = 0.150 \times 20 \times 60 = 180$ C